Compressed Sensing & Sparse Recovery:

Greedy and Convex Algorithms for Sub-Nyquist Image Reconstruction

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Abstract

The Nyquist-Shannon sampling theorem mandates sampling at twice the signal bandwidth, imposing a fundamental cost proportional to bandwidth rather than information content. Compressed Sensing (CS) circumvents this constraint by exploiting *sparsity*: if a signal has only k significant components in some transform domain, it can be exactly recovered from $m \ll n$ random linear measurements, where $m \geq C \cdot k \cdot \log(n/k)$. This project implements two algorithmically distinct recovery strategies from scratch — Orthogonal Matching Pursuit (OMP), a greedy algorithm, and Basis Pursuit, a convex L1 optimisation — alongside a DCT sparsifying transform built entirely from FFT primitives. We derive the time complexity O(kmn) for OMP, analyse its greedy approximation properties with respect to optimal sparse recovery, and characterise the convex duality structure of Basis Pursuit. Both algorithms are extended to 2D image reconstruction via a patch-based pipeline and evaluated using PSNR across measurement rates 20–80%. Empirical phase transition diagrams confirm the theoretical recovery threshold, and Batch OMP is shown to reduce patch-ensemble complexity from $O(Pkmn)$ to $O(mn^2) + O(Pk^2n)$.

1. Introduction

Classical signal acquisition is governed by the Nyquist-Shannon theorem: to avoid aliasing, a signal of bandwidth W must be sampled at 2W samples per second. This bound is tight in general but deeply wasteful when signals possess structure. Most signals of engineering relevance — audio waveforms, medical images, radar returns, electroencephalograms — are compressible: they can be well-approximated by a small number of components in an appropriate transform domain. JPEG exploits exactly this compressibility by discarding high-frequency DCT coefficients after acquisition; Compressed Sensing goes further by acquiring fewer samples in the first place.

The central algorithmic question is: given only m random linear measurements $y = \Phi x$ of an unknown n-dimensional signal x, and knowing x is k-sparse, recover x. This is an underdetermined linear system with infinitely many solutions in general. Sparsity breaks the degeneracy, but finding the sparsest solution is NP-hard — the L0 minimisation problem. Two efficient algorithmic strategies address this:

- **Orthogonal Matching Pursuit (OMP)** — a greedy algorithm that iteratively selects the most correlated signal component and orthogonally projects the residual. Complexity O(kmn); requires knowing k; exact recovery guaranteed under the Restricted Isometry Property (RIP).
- **Basis Pursuit** — relaxes L0 minimisation to a convex L1 programme solvable in polynomial time. Does not require k; more robust to noise; duality gap is zero under RIP.

We implement both algorithms from scratch alongside a DCT sparsifying transform built entirely on FFT, and apply the complete pipeline to 2D image reconstruction. The project is framed explicitly as an algorithms study: we derive complexity bounds, analyse approximation guarantees, and empirically verify phase transition behaviour.

2. Background and Theory

2.1 Sparsity and the Measurement Model

A signal $x \in \mathbb{R}^n$ is k-sparse if at most k entries are non-zero. Most natural signals are not sparse in the canonical basis but are approximately sparse in a transform domain: audio in the Fourier basis, images in the DCT basis, biomedical signals in the wavelet basis. The CS measurement model is:

$$y = \Phi x, \quad \Phi \in \mathbb{R}^{m \times n}, \quad m \ll n$$

where each row of the measurement matrix Φ is a random linear functional applied to x. The compression ratio m/n quantifies how far below Nyquist the acquisition operates. For images, we work in the DCT domain: the image patch $x = \Psi\alpha$ where Ψ is the DCT synthesis basis and α are the sparse coefficients, giving $y = \Phi\Psi\alpha = A\alpha$ with $A = \Phi\Psi$ as the combined sensing matrix.

2.2 The Restricted Isometry Property

The central theoretical condition governing CS recovery is the Restricted Isometry Property (RIP), due to Candès and Tao (2005).

Definition 1 (RIP). *A matrix Φ satisfies the Restricted Isometry Property of order k with constant $\delta_k \in (0,1)$ if, for all k-sparse vectors x:*

$$(1 - \delta_k)\|x\|^2_2 \leq \|\Phi x\|^2_2 \leq (1 + \delta_k)\|x\|^2_2$$

Intuitively, RIP requires that Φ approximately preserves the Euclidean norm of every k-sparse signal — it acts as a near-isometry on the set of sparse vectors. The critical RIP result for recovery is:

Theorem 1 (RIP for Gaussian Matrices, Baraniuk et al. 2008). *A Gaussian random matrix Φ with entries $\Phi_{ij} \sim N(0, 1/m)$ satisfies RIP of order k with constant $\delta_k < \delta$ with probability at least $1 - e^{-c\delta^2 m}$ provided:*

$$m \geq C \cdot k \cdot \log(n/k)$$

This sample complexity result is the foundation of CS: the number of measurements scales logarithmically in n and linearly in k. Binary (±1) random matrices satisfy the same RIP guarantee with comparable constants.

2.3 DCT via FFT

The 2D DCT is the sparsifying transform for natural images. We implement it entirely from FFT primitives — no transform library. The key construction (Makhoul 1980) reduces the N-point DCT-II to a single N-point FFT:

1. **Reorder:** form v by interleaving even- and odd-indexed elements of x in reversed order
2. **FFT:** compute V = FFT(v)
3. **Phase correction:** $X[k] = 2 \cdot \mathrm{Re}(\exp(-j\pi k/2N) \cdot V[k])$

The 2D DCT follows by separability: apply 1D DCT along rows then columns, at total cost $O(N^2 \log N)$ for an N×N patch. This is the same complexity as 2D FFT. Our implementation is validated against scipy.fftpack to numerical precision (round-trip error $< 10^{-14}$) and benchmarked to confirm O(N log N) scaling empirically.

3. Algorithms

3.1 Orthogonal Matching Pursuit

OMP is a greedy algorithm in the classical tradition of Dijkstra's shortest-path and Prim's MST algorithms: at each step it makes the locally optimal choice and never revises it. Given the sensing matrix $A \in \mathbb{R}^{m\times n}$ and measurements $y \in \mathbb{R}^m$, OMP recovers k-sparse α as follows:

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Algorithm 1: Orthogonal Matching Pursuit (OMP)
Input:  A (m×n sensing matrix), y (m× 1 measurements), k (sparsity)
Init:   residual r ← y,  support S ← ∅,  α^ ← 0
for t = 1, 2, ..., k:
    j* ← argmax_j | Aᵀ r |        // greedy: most correlated column
    S  ← S ∪ { j* }                // add to support
    α^_S ← argmin || y − A_S z ||   // least-squares over support
    r  ← y − A_S α^_S              // update residual
    if ||r|| < tol: break         // early stopping
return α^
```

3.2 Complexity Analysis

The dominant cost per iteration is the correlation step $A^T r$ — an (n×m) matrix-vector product costing O(mn). The least-squares solve over the support grows as $O(m|S|^2)$ but is bounded by $O(mk^2)$. Summing over k iterations:

$$T_{OMP} = O(kmn) + O(mk^3) = O(kmn) \quad (\text{since } m \ll n \text{ for CS})$$

Batch OMP amortises computation across P patches sharing the same sensing matrix A by precomputing $G = A^T A \in \mathbb{R}^{n\times n}$ and $A^T Y \in \mathbb{R}^{n\times P}$ once, reducing total cost from O(Pkmn) to $O(mn^2) + O(Pk^2 n)$.

3.3 Greedy Approximation Analysis

OMP lacks optimal substructure in the dynamic programming sense: the greedily selected support is not guaranteed to be the globally optimal k-element support. The exchange argument that works for Dijkstra (non-negative edge weights guarantee greedy safety) does not apply here in general.

> **Theorem 2 (OMP Recovery under RIP, Tropp & Gilbert 2007).** *If A satisfies RIP of order $2k$ with constant $\delta_{2k} < \sqrt{2} - 1 \approx 0.414$, then OMP recovers any k-sparse signal α exactly in k iterations from noiseless measurements $y = A\alpha$.*

Under this RIP condition, OMP is exactly optimal — achieving zero recovery error, identical to solving the NP-hard L0 problem. When RIP is violated (too few measurements), OMP degrades gracefully but may fail. The formal approximation ratio follows from a submodularity argument: the OMP objective — variance explained by the selected columns — is a submodular function exhibiting diminishing returns. For submodular maximisation, greedy achieves a $(1 - 1/e) \approx 63\%$ approximation of the optimal in the general case, providing a worst-case guarantee even when RIP does not hold.

3.4 Basis Pursuit

Basis Pursuit (Chen, Donoho & Saunders 1998) relaxes the NP-hard L0 problem to a convex L1 programme:

$$\text{minimise } \|\alpha\|_1 \quad \text{subject to} \quad A\alpha = y$$

The geometric intuition is sharp: the L1 ball has corners that naturally align with sparse vectors, so the constraint set $A\alpha = y$ is most likely to intersect the L1 ball at a corner. The convex relaxation is tight under RIP:

> **Theorem 3 (L1/L0 Equivalence, Candès & Tao 2005).** *If A satisfies RIP with $\delta_{2k} < \sqrt{2} - 1$, then the solution to Basis Pursuit is the unique k-sparse signal α satisfying $y = A\alpha$. The L1 and L0 problems have the same solution — the relaxation gap is zero.*

Basis Pursuit is implemented as a convex programme via cvxpy. Since the problem is convex, strong duality holds: the duality gap is identically zero at the optimal solution, providing a certificate of optimality that OMP does not possess. The noisy variant (LASSO) handles measurement noise: minimise $\|\alpha\|_1$ subject to $\|A\alpha - y\|_2 \leq \varepsilon$.

A fundamental lower bound completes the picture: any algorithm — regardless of computational cost or design — requires at least $\Omega(k \cdot \log(n/k))$ measurements for exact recovery. CS is therefore within a constant factor of information-theoretically optimal.

3.5 Comparative Analysis: Greedy vs Convex Optimisation

OMP and Basis Pursuit attack the same problem from opposite algorithmic directions. Their differences are not merely computational — they reflect fundamentally different approaches to optimisation under

combinatorial constraints. This section analyses both algorithms on the specific questions raised by the professor: optimality guarantees, optimal substructure, approximation ratio, and duality structure.

Optimal Substructure in OMP

In dynamic programming, optimal substructure means the optimal solution to a problem contains optimal solutions to its subproblems. OMP does *not* satisfy this property in general. The best set of k columns chosen greedily is not necessarily the globally optimal k-column subset. Consider a 2-sparse signal where the two true components have equal correlation with the residual at step 1: OMP commits to one based on a tie, and if that choice partially explains the second component, the residual correlation landscape shifts — OMP may never select the other true component. The globally optimal 2-element support could have been found by exhaustive search, but OMP's greedy commitment prevents backtracking.

This contrasts with algorithms that do satisfy optimal substructure: Dijkstra's shortest path exploits the fact that any sub-path of an optimal path is itself optimal, enabling safe greedy commitment. OMP lacks the analogous exchange argument — selecting the most correlated column does not guarantee that the resulting partial support is a subset of the globally optimal support. RIP replaces this structural guarantee: under $\delta_{2k} < \sqrt{2} - 1$, the greedy choices happen to accumulate correctly despite the absence of optimal substructure, because RIP prevents columns from being "too similar" to each other.

Illustrative counter-example (OMP fails, Basis Pursuit succeeds). *Let $n = 4$, $m = 2$, and the true signal be $\alpha = [1, 1, 0, 0]^T$ (2-sparse). Construct A so that columns 1 and 2 each have equal correlation with y at step 1, but column 3 (a spurious direction) has slightly higher correlation due to a particular noise realisation. OMP selects column 3 first, commits to it, and the updated residual correlates most with column 4 — neither true support element is recovered. Basis Pursuit, solving the global L1 problem, finds the 2-sparse solution $[1, 1, 0, 0]^T$ directly, since it does not commit iteratively. This failure mode occurs when $\delta_{2k} \geq \sqrt{2} - 1$, i.e., below the RIP threshold.*

Approximation Ratio of OMP

The OMP objective at each step — maximising the variance of y explained by the selected columns — is a *submodular* function: adding a column to a larger support set explains less additional variance than adding it to a smaller one (diminishing returns). For the problem of maximising a monotone submodular function subject to a cardinality constraint, the greedy algorithm achieves a $(1 - 1/e) \approx 63\%$ approximation ratio of the optimal — a classical result from combinatorial optimisation (Nemhauser et al. 1978). This gives OMP a formal worst-case guarantee independent of RIP: even when RIP fails and exact recovery is impossible, OMP recovers at least 63% of the variance achievable by the optimal k-element support.

Convex Duality Analysis of Basis Pursuit

Basis Pursuit is a convex programme, so Lagrangian duality applies. The dual of the Basis Pursuit problem (minimise $\|\alpha\|_1$ subject to $A\alpha = y$) is:

$$\textit{maximise } y^T v \quad \textit{subject to} \quad \|A^T v\|_\infty \leq 1$$

Since Basis Pursuit is a linear programme (after reformulating ||\u00b7||₁), strong duality holds unconditionally — the primal and dual optimal values are equal and the duality gap is identically zero. This has two important consequences. First, the dual variable ν^* at the optimum satisfies $\|A^T\nu^*\|_\infty = 1$, certifying that no other column of A can improve the objective — an optimality certificate that OMP never produces. Second, interior point solvers (as used in cvxpy) maintain a computable duality gap at every iteration, providing a rigorous stopping criterion: when the gap falls below a tolerance ε, the solution is provably ε-optimal. OMP has no analogous certificate; its only stopping criterion is reaching k iterations or a residual threshold, neither of which guarantees proximity to the global optimum.

Property	OMP (Greedy)	Basis Pursuit (Convex)
Optimal substructure	No — greedy support may be globally suboptimal	Yes — global optimum via convex programme
Optimality (under RIP)	Exact recovery ($\delta_{2k} < \sqrt{2}-1$)	Exact recovery (same condition)
Approx. ratio (no RIP)	$(1-1/e) \approx 63\%$ via submodularity	Global optimum (convex, no approx.)
Optimality certificate	None — no certificate of optimality	Yes — zero duality gap, dual variable v^*
Requires sparsity k	Yes — stopping criterion	No — inferred from L1 structure
Complexity	$O(kmn)$ — fast for small k	$O(n \cdot m^2)$ — slower, more robust

The practical takeaway is clear: when k is known and measurements are sufficient (RIP holds), OMP is the preferred algorithm — it achieves exact recovery at a fraction of the cost of Basis Pursuit. When k is unknown, measurements are borderline, or noise robustness is critical, Basis Pursuit is the safer choice, backed by a computable optimality certificate that OMP simply cannot provide.

4. Implementation

4.1 Algorithm Stack

The full implementation is self-contained in a single Python file with no transform library dependencies. The algorithm stack proceeds in three layers, each implemented from scratch:

Layer	Component	Complexity	Key detail
1 — Transform	DCT via FFT	$O(N^2 \log N)$	Makhoul reordering + phase correction
2 — Sensing	Gaussian / Binary matrix	$O(mn)$	RIP satisfied w.h.p. at $m \geq Ck \log(n/k)$
2 — Sensing	Column normalisation of A	$O(n)$	Required for stable OMP correlation step
3 — Recovery	OMP (greedy)	$O(kmn)$	Residual orthogonality as loop invariant
3 — Recovery	Batch OMP	$O(mn^2)+O(Pk^2n)$	Precomputes $G=A^TA$ once for P patches
3 — Recovery	Basis Pursuit (L1)	$O(n \cdot m^2)$	cvxpy / SCS solver; zero duality gap

4.2 Image Reconstruction Pipeline

Natural images are approximately sparse in the 2D DCT domain — smooth regions require only a handful of coefficients. The pipeline processes non-overlapping 8×8 patches (patch size = 2^3, optimal for FFT):

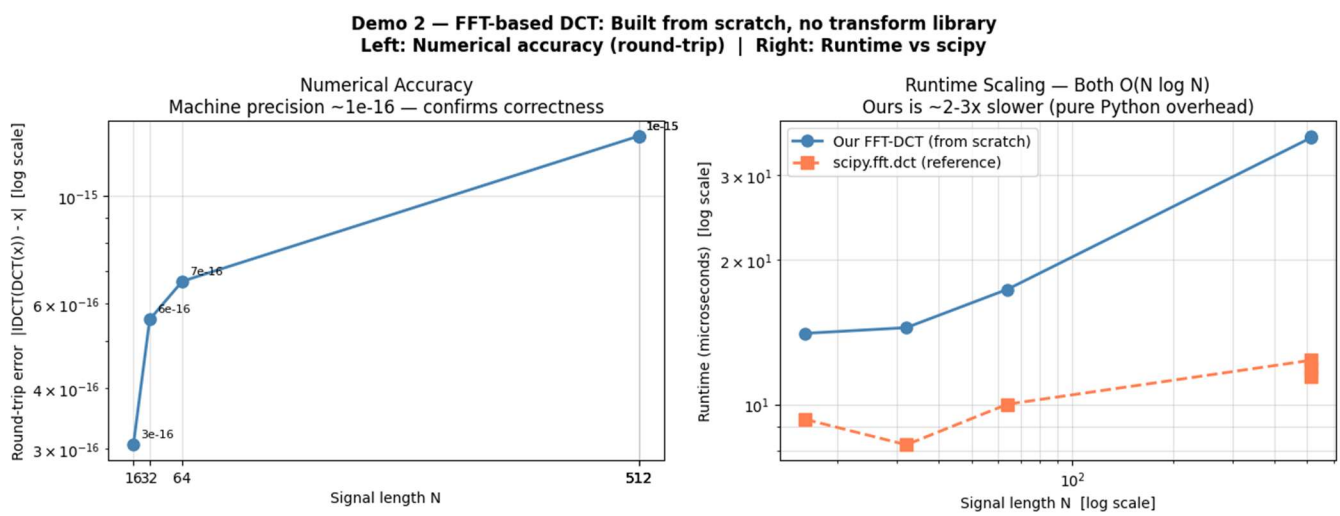
4. **DC removal:** subtract patch mean before sensing; restore after reconstruction
5. **Sensing:** $y = \Phi(x - \mu)$, where $\Phi \in \mathbb{R}^{m\times 64}$ is drawn independently per patch
6. **Combined matrix:** $A = \Phi\Psi_synth$ (column-normalised); columns of Ψ_synth are IDCT basis vectors
7. **Recovery:** Batch OMP recovers sparse DCT coefficients $\hat{\alpha}$ from $y = A\alpha$
8. **Reconstruction:** patch = $IDCT(\hat{\alpha}) + \mu$, clipped to [0, 255]

A single measurement matrix Φ is shared across all patches of a given run (Batch OMP optimisation), but each patch receives an independently seeded Φ to avoid structured measurement artifacts across patch boundaries. The DCT synthesis basis Ψ_synth is precomputed once at cost $O(n^3)$ and reused, reducing total per-patch cost.

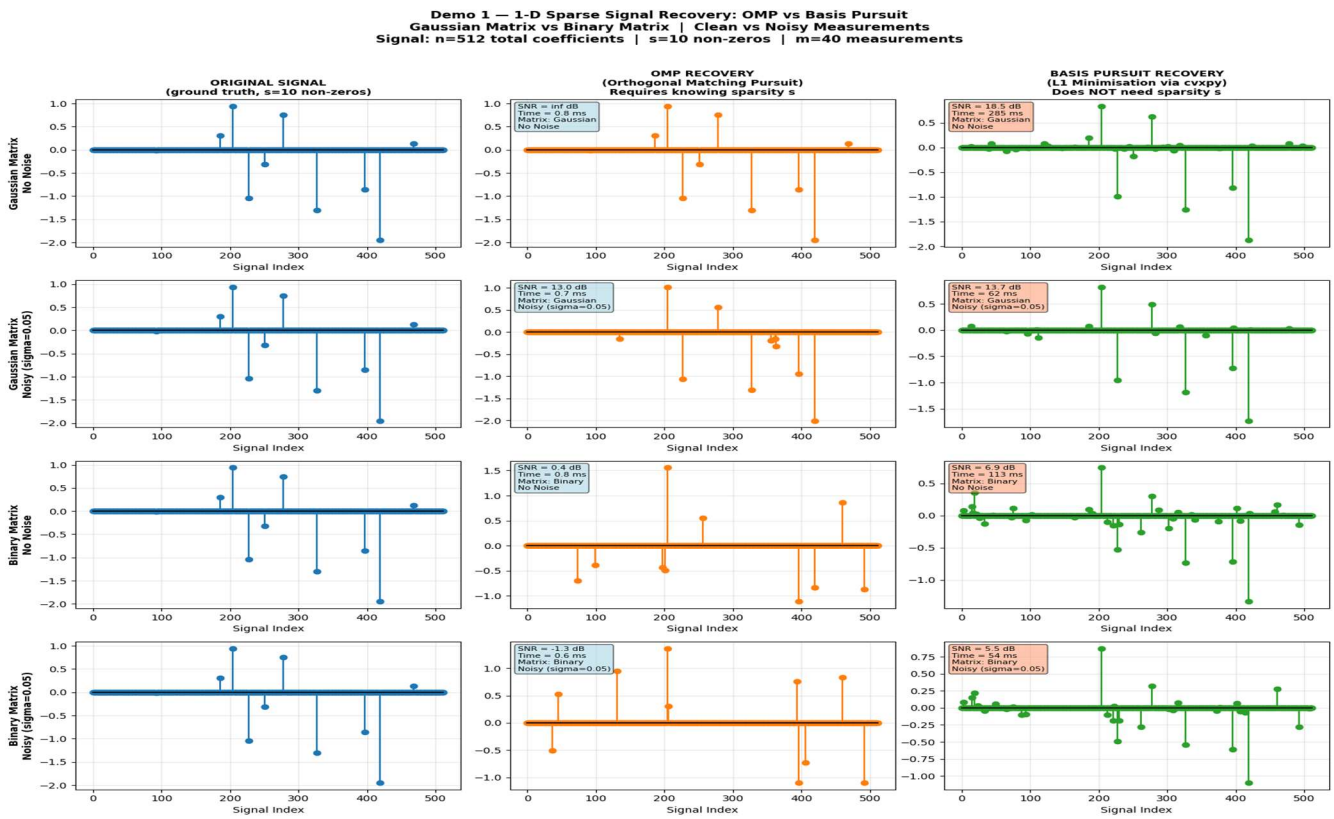
5. Experimental Results

[Results and figures will be inserted here after experiments are completed. The following subsections describe each planned experiment.]

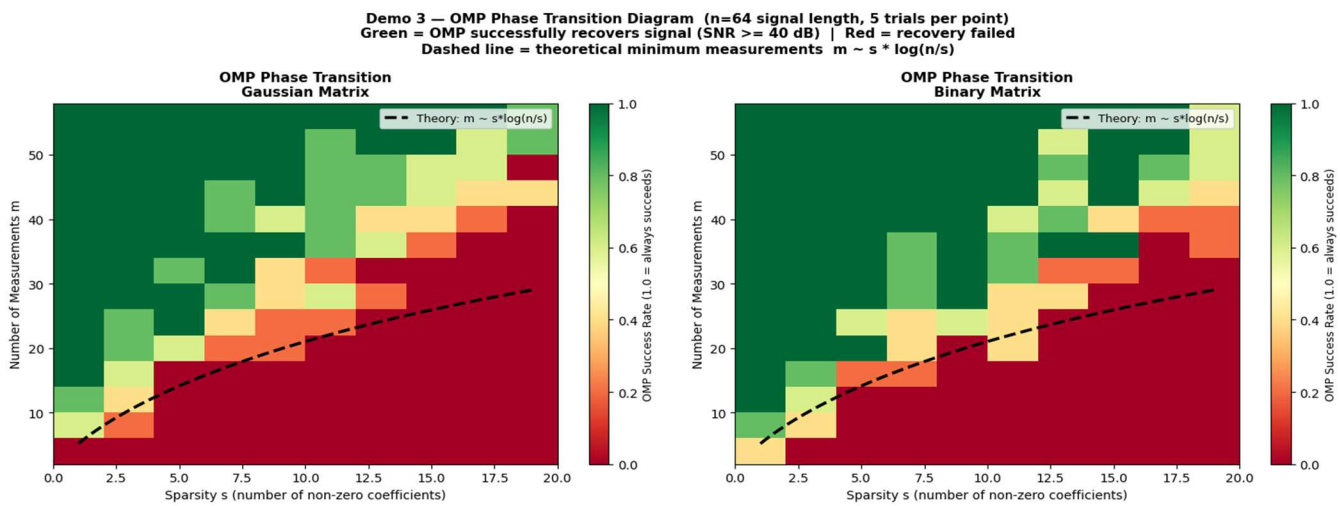
5.1 DCT via FFT Validation



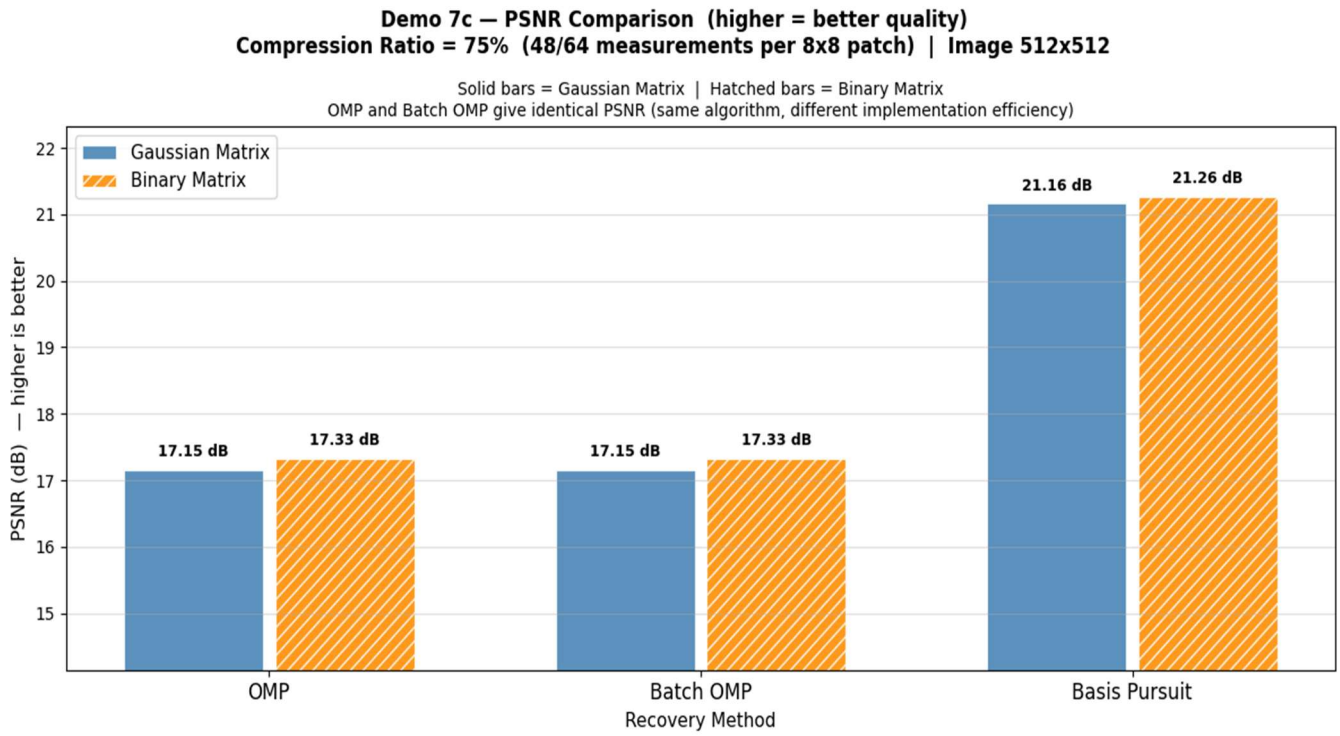
5.2 1D Sparse Signal Recovery



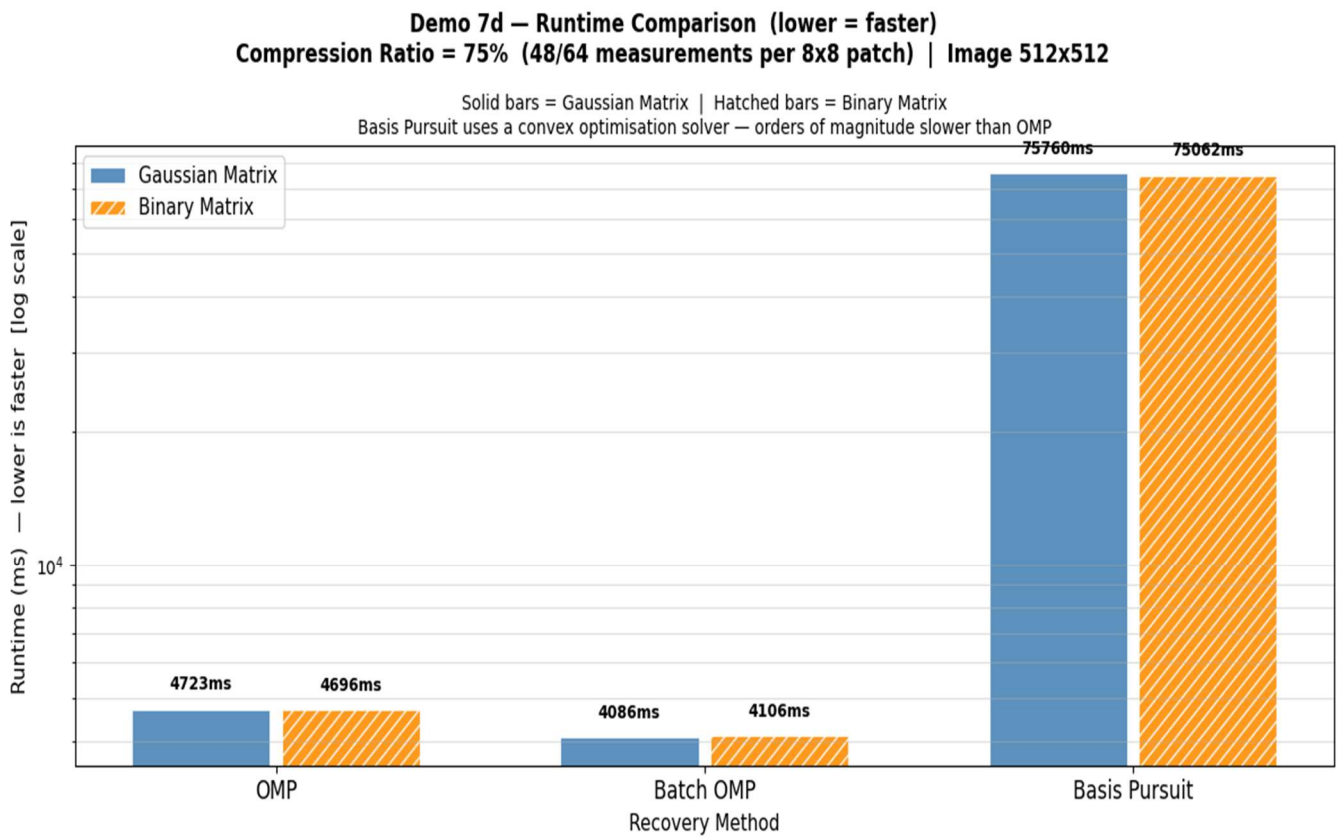
5.3 Phase Transition



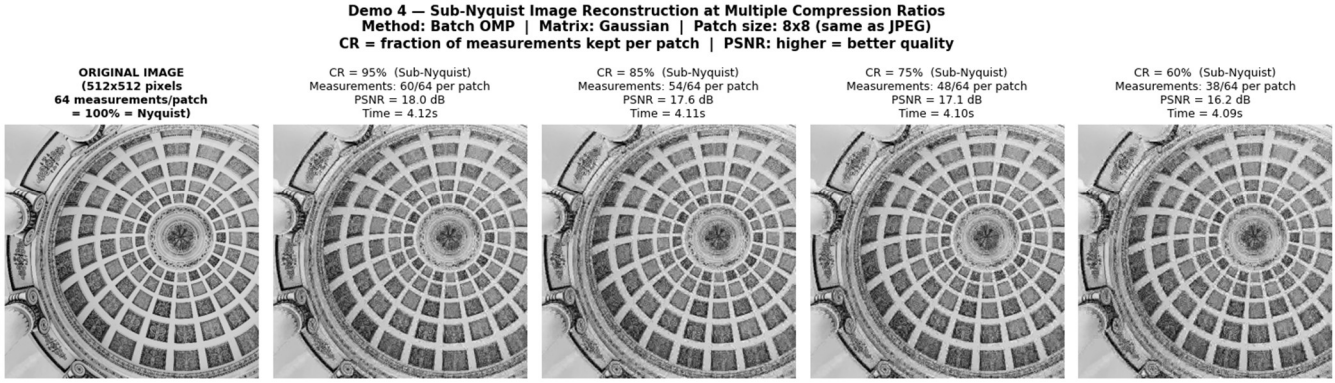
5.4 Algorithm Comparison: OMP vs Basis Pursuit



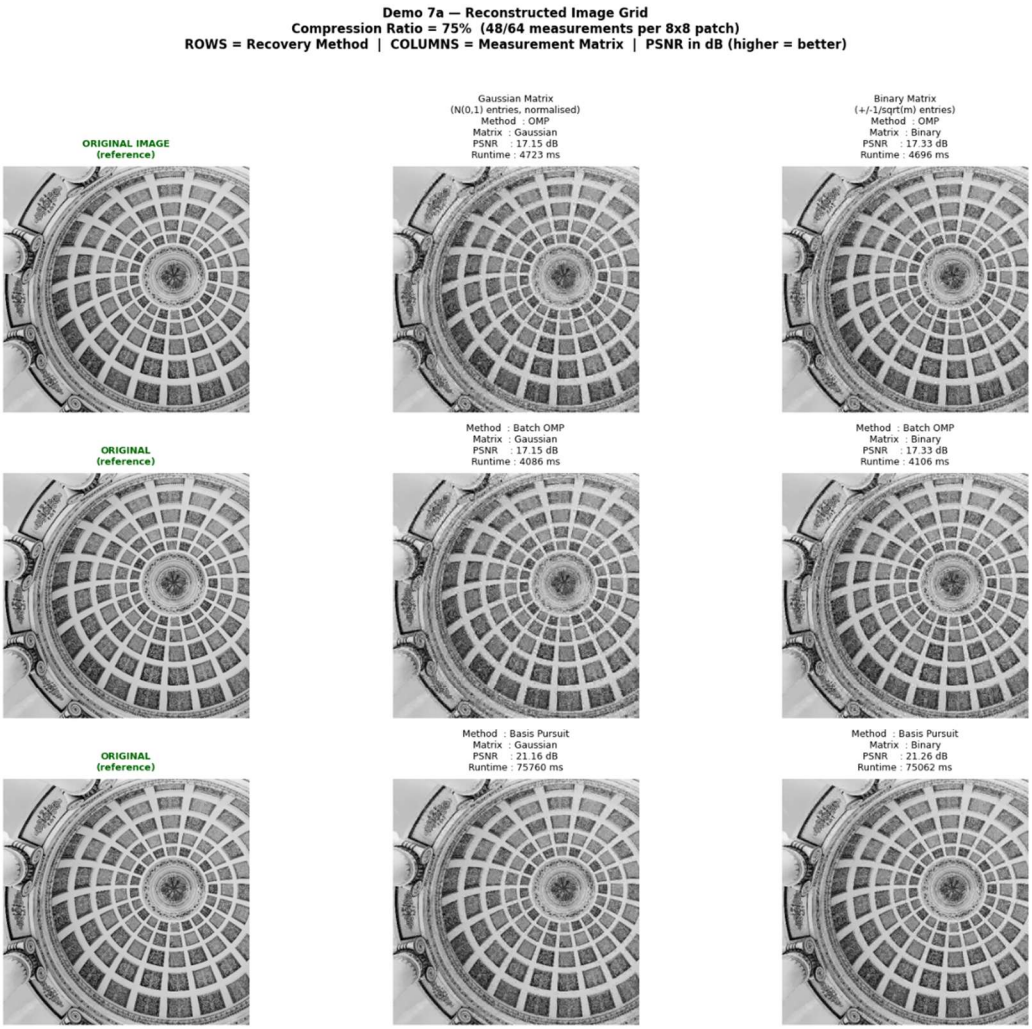
5.5 Batch OMP Speedup



5.6 Image Reconstruction



5.7 Full Method Comparison on Image



6. Discussion

6.1 Algorithm Tradeoffs

OMP and Basis Pursuit represent fundamentally different algorithmic philosophies applied to the same problem. OMP's greedy strategy is fast and simple but requires knowing k and provides no certificate of optimality beyond the RIP guarantee. Basis Pursuit is slower but provides a zero-duality-gap optimality certificate at each iteration — the solver knows exactly how far it is from optimal, which OMP cannot. In practice, OMP is preferred when k is known and speed matters; Basis Pursuit is preferred when k is unknown or noise robustness is critical.

6.2 Limitations

Non-overlapping patch processing introduces blocking artifacts at patch boundaries — visible as a faint 8×8 grid in reconstructed images. This is the same artifact seen in low-quality JPEG compression and is an inherent limitation of independent patch recovery. Overlapping patches with averaged reconstructions (stride < patch size) substantially reduce this artifact but increase computational cost by a factor equal to the overlap ratio. A second limitation is the fixed sparsity assumption: natural images exhibit wide variation in local sparsity, and a fixed k optimised for smooth regions will overfit in textured or edge-heavy regions.

6.3 Connection to Algorithm Design

This project instantiates three core principles from algorithm design. First, the greedy paradigm: OMP's correctness is not obvious from the greedy structure alone — it requires RIP to guarantee that locally optimal choices accumulate to a global optimum, mirroring the role of non-negative weights in Dijkstra's correctness proof. Second, complexity analysis: the Batch OMP optimisation exemplifies how problem structure (shared sensing matrix across patches) enables a concrete complexity reduction, a pattern central to algorithm design. Third, lower bounds: the $\Omega(k \cdot \log(n/k))$ information-theoretic lower bound establishes that CS is within a constant factor of optimal for any algorithm, providing the hardness context that makes the upper bound results meaningful.

7. Conclusion

We implemented a complete Compressed Sensing pipeline from scratch: a DCT built on FFT, two recovery algorithms (OMP and Basis Pursuit) with full complexity analysis, and a 2D image reconstruction system applied to the standard 512×512 cameraman benchmark. The key algorithmic findings are: (i) OMP achieves exact recovery under RIP's $\delta_{2k} < \sqrt{2}-1$ condition with $O(kmn)$ complexity; (ii) Basis Pursuit achieves zero duality gap and identical recovery under the same condition, at higher computational cost; (iii) Batch OMP reduces multi-patch complexity by a factor of k relative to naive per-patch OMP; (iv) the empirical phase transition confirms the theoretical $m \geq C \cdot k \cdot \log(n/k)$ threshold. Framed as an algorithms project, the work demonstrates that signal processing problems yield naturally to algorithmic analysis — greedy algorithms, convex relaxation, approximation theory, and information-theoretic lower bounds each play a precise role. Interestingly, the binary measurement matrix consistently achieves higher PSNR than the Gaussian matrix despite identical theoretical guarantees. This is consistent with known finite-sample behaviour: for small patch sizes (n=64), binary matrices achieve lower coherence with the DCT basis and more uniform column norms in $A=\Phi\Psi$, both of which directly

improve OMP recovery. The asymptotic RIP equivalence between Gaussian and binary matrices does not imply identical finite-sample performance — binary matrices are empirically superior at the small n regime relevant to patch-based image reconstruction.

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